The Second Corpus Study on German Nominal Compounding Constraints

Background

German nominal compounding is a notoriously unpredictable and irregular process. About 40% of German N+N compounds insert a so-called linking element that is not confidently predictable for most of constituent classes. For example, nouns Birne ‘pear’ and Kirsche ‘cherry’ build compounds Birne-n-saft ‘pear juice’ but Kirsch-0-saft ‘cherry juice’ despite having nearly identical phonological, morphological, and semantic properties. The only linguistic generalizations available to gauge the correct linker are (mostly) irregular constraints that are eagerly violated by numerous compounds.

Our first corpus study on German nominal compounding collected 64 constraints from the linguistic literature and verified 39 of them against ~280k German N+N compounds. The corpus study demonstrated that most of the constraints were irregular, indeed. The most crucial funding was that not all of the constraints were confirmed on the massive data. Specifically, eight constraints were only partially supported by the data, and three were not supported at all.

The main limitation of the first corpus study was that none of the constraints involving semantic or lexical conditions could be implemented. The present study fills in most of the gaps remaining after the first corpus study:

1. It implements 17 constraints involving semantics. Fourteen of the 17 constraints are pluralic constraints actively discussed in recent linguistics literature.
2. It applies more detailed statistic analyses to analyze relations between different ontological properties of the constraints (application manner, types of linguistic conditions etc.) and its coverage and regularity measures.
3. It additionally analyses constraints through the Yang 2016's Tolerance Principle lense and makes a first attempt to discriminate between linguistic constraints and statistical patterns, and to model compounding with the Elsewhere Condition.

Additionally, it implements a couple of further constraints and verifies the constraints on further datasets.

Datasets

GeCoDB_v06

> 282767 compound lemmata. 3614 N1 lemmata. 9 linkers, 11 linker allomorphs. 4768 unique N1+linker combinations.

~280k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Database/compounds/decow16ax_comps_counts.txt'), with further processing steps in Shmalts BA Thesis, final preprocessing in Shmalts 2026.

The dataset is used to (re-)verify 40 non-semantic constraints.

GeCoDB_v07

> 275411 compound lemmata. 3593 N1 lemmata. 10 linkers, 12 linker allomorphs. 4741 unique N1+linker combinations.

~275k German N+N compounds from *GeCoDB_v06* whose constituents are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06*.

GeCoDB_v07_GN

> 48976 compound lemmata (48660 unique N1+N2 combinations). 3319 N1 lemmata. 10 linkers, 12 linker allomorphs. 3585 unique N1+linker combinations.

~49k German N+N compounds from *GeCoDB_v07* that are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06* and five further semantic non-pluralic constraints.

SP18_Pl_Constr

> 2681 compound lemmata. 44 N1 lemmata. 8 linkers, 9 linker allomorphs. 78 unique N1+(Sg/Pl meaning)

> combinations. 88 unique N1+linker combinations.

~2.5k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo, with annotated singular/plural meaning of N1. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Corpusstudy/Concordance.full.csv').

The dataset is used to verify 12 pluralic constraints.

Outline

1. We first inspect the ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language: application manner, type(s) of accessed linguistic properties, application item. The results corroborate and slightly elaborate the finding from the first corpus study.
2. We then inspect the impact of data and metric choice on surface constraint applicability. We find that the constraint applicability values are statistically very similar both across the *GeCoDB* dataset versions and the item/type variants of the metrics. Based on these results, we focus on the results of *GeCoDB_v07_GN* and *SP18_Pl_Constr* and only concentrate on the item variant of the applicability statistics.
3. We analyze the values of application measures of the constraints on the flattened applicability statistics, which includes their distribution and correlation. We establish relation between the values of coverage and regularity of the constraints, and their ontological properties from step 1.
4. We finally introduce and apply the new (irregularity) scale to our results.

Results

1. Constraint Ontological Properties

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. Al but two out of 65 constraints do not access the linguistic properties of N1.

2. Constraint Applicability Vs. Datasets & Metrics

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated across all pairwise comparisons of *GeCoDB* versions. That speaks in favor of the validity of the original results. Additionally, it attests the sufficient data quality and ensures the robustness of the constraint applicability statistics. We choose to proceed with *GeCoDB_v07_GN* as the dataset with the most verified constraints.
- The item and type variants of the applicability metrics are statistically very similar, which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

3. Constraint Applicability

- Most of the item coverage values are very low at <0.05. The item regularity values are more even. They are clustered naturally by the thresholds of 0.7 and 0.9, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$, $[0.7; 0.9)$, $[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints, while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho -0.184 and -0.342, respectively).
- We find that including semantic conditions to constraint definitions statistically sinks the item regularity of the constraints (coeff -0.21 at intercept 0.85, p-value 0.001), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

4. Constraint Support, Productivity & Regularity

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 are now partially supported. Two out of five new constrains verified by *GeCoDB_v07_GN* are not

supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*.
- Eleven constraints out of 49 turn out to be unproductive by the Tolerance Principle. Regular constraints by the RTP constitute slightly less than a half of the constraint set: 23 out of 49.
- We find that no constraints require changes in their conditions. Only regularity levels should be adjusted for them.

1. Constraint Ontological Properties

Description

In this chapter, we are interested in ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language. More specifically, we will inspect the following properties of the constraint set:

1. The distribution of the constraints by the application manner.
2. The distribution of the constraints by the type(s) of accessed linguistic properties (domain type). Note that the default constraint is excluded from consideration in this step.
3. The distribution of the constraints by the application item.

NB! In this chapter, we first take a look at the entire set of constraints, whether they are confirmed or not.

Results

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. All but two out of 65 constraints do not access the linguistic properties of N1.

1. In total, 65 constraints are summarized in this corpus study. P2L constraints make almost double of L2P constraints.

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	N
total	65
p2l	42
l2p	22

2. We can clearly see that the morphological (including mixed) constraints constitute the core of the constraint set, making almost a quarter of all constraints. If we sum up all constraint involving morphology (so mixed as well), this number rises to $>50\%$>50\%$. Derivational and phonological constraints occupy a substantial share of the set as well. Semantics is negligible when isolated but makes 14.1%14.1\%$ of all constraints when coupled with morphology. Lexical and further mixed constraints are overall rare. The same distribution holds as well for the P2L and the L2P subsets of constraints, except for lexical constraints being the second most frequent type of L2P constraints, while the number of lexical P2L constraints is one.

The drop in domain type frequency is overall quite gradual, with no large gaps: 18.8%18.8\%$, 17.2%17.2\%$, 14.1%14.1\%$, 9.4%9.4\%$, then 7.8%7.8\%$, etc. Notably, the constraints are distributed much more evenly by the linguistic domain type among the P2L constraints than among the L2P constraints. The P2L subset resembles the entire constraitn set in this regard, with a gradual decrease in domain type frequency. On the contrary, domain type frequency quickly drops to low values in the L2P subset, whose top three domain types (morphological, lexical, make derivational+phonological constraints) make 13, or almost 60%60\%$, of all L2P constraints. These are then followed by a long tail of underrepresented types.

The L2P constraints almost ignore semantics. That means that German nominal compounding does not have formal semantic restrictions that would prohibit some linkers from appearing in certain semantic contexts (as opposed to, say, morphological formal restrictions). The P2L constraints, on the contrary, draw a lot on semantics. This, and the fact that most of the semantics-involving constraints are coupled with a morphological condition, indicates that semantics only has an complementary function in German nominal compounding. It can only serve as a soft clue when generating compounds but cannot really restrict linker choice.

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	total		p2l		l2p	
	N	%	N	%	N	%
morphological	12	18.8	6	14.6	6	28.6
derivational	11	17.2	9	22.0	2	9.5
morphological+semantic	9	14.1	8	19.5	1	4.8
phonological	6	9.4	5	12.2	1	4.8
morphological+phonological+semantic	5	7.8	4	9.8	1	4.8
lexical	5	7.8	1	2.4	4	19.0
morphological+phonological	3	4.7	2	4.9	1	4.8
derivational+phonological	3	4.7	0	0.0	3	14.3
derivational+morphological	2	3.1	1	2.4	1	4.8
derivational+semantic	2	3.1	2	4.9	0	0.0
semantic	2	3.1	2	4.9	0	0.0
derivational+morphological+phonological	1	1.6	1	2.4	0	0.0
lexical+semantic	1	1.6	1	2.4	0	0.0
lexical+morphological+phonological+semantic	1	1.6	0	0.0	1	4.8
lexical+morphological	1	1.6	0	0.0	1	4.8

The conclusions made above are supported by the individual contributions of linguistic domains. By the individual contribution of a domain, we mean the total number of constraints that access linguistic properties from this domain, both mixed or not. For example, the individual contributions in a small set of three constraints [morphological, mophological+phonological, derivational+phonological] are: morphology: 2, phonology: 2, derivation: 1.

We can clearly see the overwhelming contribution of morphology. Slightly more that half of all constraints have a morphological condition. Semantics, derivation, and morphology have almost equal contributions, being present in almost a third of constraints each.

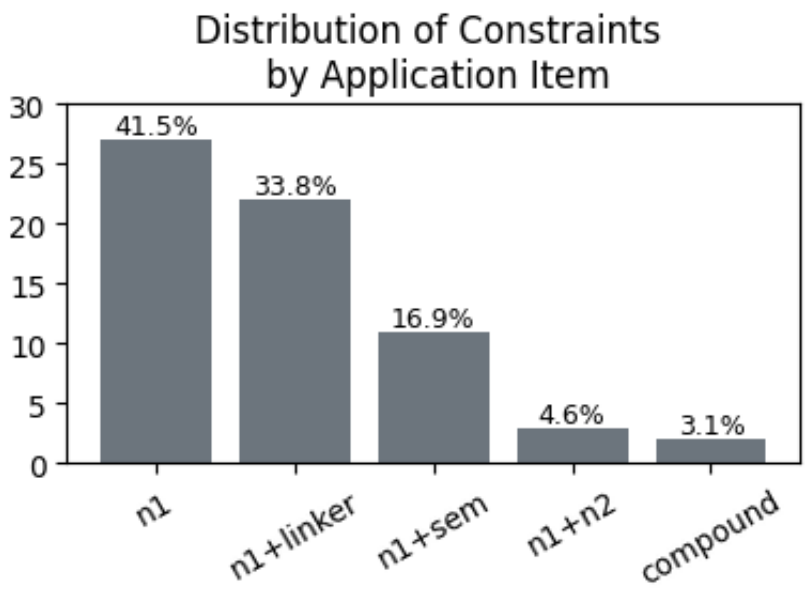
While the counts for unique domain types among the P2L and the L2P differs significantly, the individual contribution of each considered linguistic domain in not that different. Thus, morphology, phonology, and derivation contribute almost equally to P2L and L2P constraints. However, lexical constraints are prevalent among the L2P constraints, and semantic ones -- among the P2L constraints, as discussed above. For instance, semantic constraint contribute to $>40\%$$>40\%$ of the P2L but only to $14.3\%$$14.3\%$ of the L2P constraints. The absolute numbers are even worse: $17$$17$ vs $3$$3$.

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	total		p2l		l2p	
	N	%	N	%	N	%
morphological	34	53.1	22	53.7	12	57.1
semantic	20	31.2	17	41.5	3	14.3
derivational	19	29.7	13	31.7	6	28.6
phonological	19	29.7	12	29.3	7	33.3
lexical	8	12.5	2	4.9	6	28.6

3. N1 plays the most prominent role in constraint definitions. Being the most frequent type when isolated, it also constitutes part of the further most frequent items: N1 + linker, N1 + plural semantics, etc. Actually, only two out of 65 constraints do not access the properties of N1.

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Below are the counts of unique items for each item type in each dataset.

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	compound	n1	n1+sem	n1+n2	n1+linker
gecodb_v06	282767	3614	0	0	4870
gecodb_v07	275411	3593	0	0	4741
gecodb_v07_gn	48976	3319	0	48660	3585
sp18_pl_constr	0	0	78	0	88

2. Constraint Applicability Vs. Datasets & Metrics

Description

Before we proceed to the analysis of coverage and regularity of the constraints, we need to inspect the impact of data and metric choice on surface constraint applicability.

1. We want to understand how robust the findings of the first corpus study are, and whether the (unexpected) irregularity of some constraints also extends to further dataset and is not a consequence of data choice.
2. We evaluate the difference between the item and type variant of coverage and regularity.

To assess that, we will run some statistical tests over the results obtained across different versions of the *GeCoDB* dataset, and across the item/type variant of applicability metrics.

NB! In this chapter, we take a look at the entire set of constraints, whether they are confirmed or not.

Results

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated across all pairwise comparisons of *GeCoDB* versions. That speaks in favor of the validity of the original results. Additionally, it attests the sufficient data quality and ensures the robustness of the constraint applicability statistics. We choose to proceed with *GeCoDB_v07_GN* as the dataset with the most verified constraints.
- The item and type variants of the applicability metrics are statistically very similar, which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

Out of 65 constraints, we have the applicability statistics for 54. The application manner of those is nearly identical to that of the entire constraint set: the P2L constraints almost double the L2P subset. From the missing 11 constraints, three had no covered compounds in the datasets (one in *GeCoDB*'s and two in *SP18_Pl_Constr*), and the remaining eight are not implemented. These all include semantic or lexical conditions.

GeCoDB_v07_GN covers the most constraints (44). Also note that since all pluralic constraints are only verified over *SP18_Pl_Constr*, there are no overlap between this dataset and the *GeCoDB*'s.

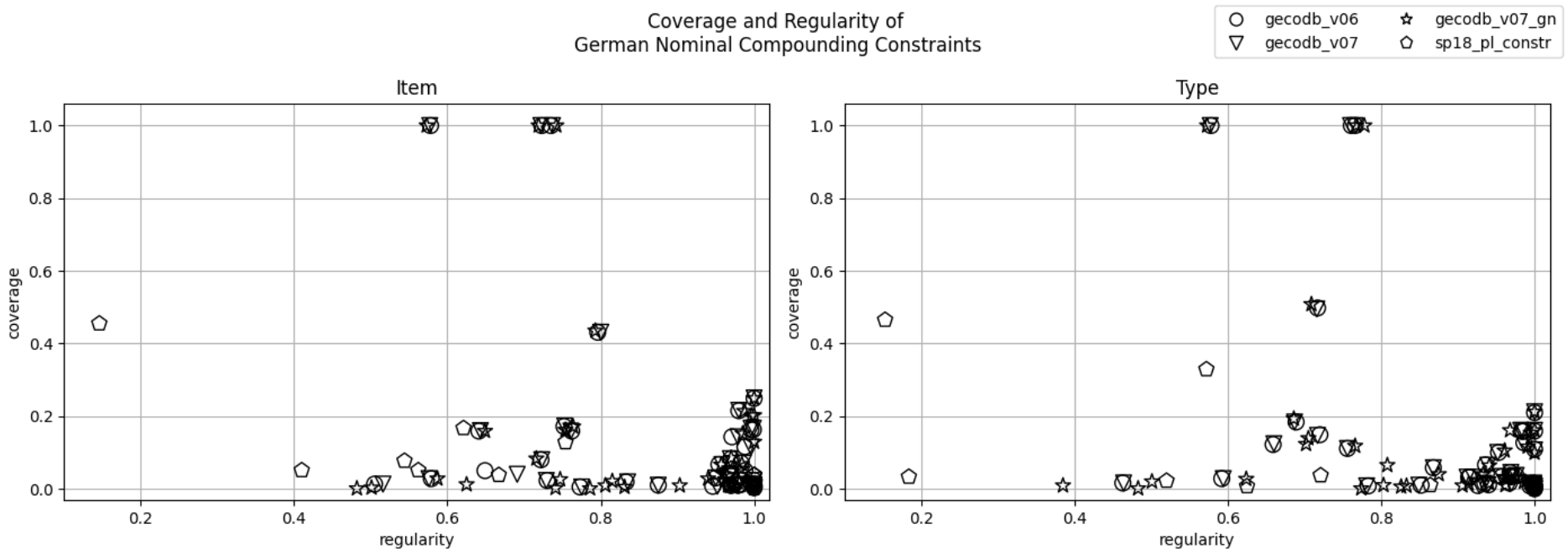
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	total	gecodb_v06	gecodb_v07	gecodb_v07_gn	sp18_pl_constr
total	54	39	39	44	10
p2l	36	23	23	27	9
l2p	17	15	15	16	1

Draw the plots of item and type coverage and regularity values across the datasets and metrics.

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1. One can clealy see that almost all the values from different versions of the *GeCoDB* dataset are located at the same locations. For each unique pair of these dataset versions, we run a Wilcoxon signed-rank test to test whether the values of item and type coverage and regularity across the pair are at the same location (we consider only shared constraints, since *GeCoDB_v07_GN* verifies a few more). We choose the Wilcoxon signed-rank test because we are dealing with paired samples that are distributed abnormally. We additionally calculate the Spearman correlation coefficient to test for the mutual predictability of the values. We choose Spearman because, again, our values are distributed abnormally.

The results reveal that none of the comparisons reach a significant result of the Wilcoxon signed-rank (all p-values $>0.125$$>0.125$; two-sided alternative hypothesis: the locations are shifted). Moreover, the Spearman correlation coefficient is in a range of $[0.946; 1.0]$$[0.946; 1.0]$ for all comparisons. We conclude that the results from different *GeCoDB* versions are interchangeable, and choose to proceed with *GeCoDB_v07_GN* as the dataset with the most of verified constraints.

The fact the results from *GeCoDB_v07_GN* are statistically identical to the results of *GeCoDB_v06* and *GeCoDB_v07* provide additional evidence that the earlier *GeCoDB* versions were and are sufficiently clean and eligible for the corpus study, and that the results of the first corpus study are valid and legitimate. While one might have argued that unexpectedly low regularity of some constraints found in Shmalts 2026 was a result of the presence of low-quality data from the web in *GeCoDB_v06* (typos, unconventional compounds, etc.), compounds from GermaNet, that produce the same results, can hardly be accused of that.

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		reg		cvg	
		item	type	item	type
comparison					
gecodb_v06 vs. gecodb_v07	W	258.500	247.000	142.000	171.500
	p	0.257	0.474	0.577	0.472
	r	0.988	0.999	0.999	1.000
gecodb_v06 vs. gecodb_v07_gn	W	273.500	274.000	273.000	355.000
	p	0.138	0.140	0.268	0.675
	r	0.946	0.986	0.972	0.990
gecodb_v07 vs. gecodb_v07_gn	W	272.500	270.000	272.000	344.000
	p	0.134	0.126	0.202	0.694
	r	0.952	0.988	0.972	0.990

2. The comparison of the item and type variants of the metrics allows to conclude that the two variants are statistically very similar. We choose to proceed with the item variant as the more balanced version of the metrics.

First, the central tendency measures between the two variants are almost identical for both coverage and regularity.

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		count	mean	std	min	50%	90%	max
reg	item	54.0	0.813	0.220	0.088	0.922	1.000	1.0
	type	54.0	0.826	0.212	0.153	0.924	1.000	1.0
cvg	item	54.0	0.125	0.236	0.001	0.037	0.239	1.0
	type	54.0	0.121	0.240	0.000	0.028	0.294	1.0

Furthermore, the results of the Wilcoxon signed-rank test suggest that the two variants of both coverage and regularity come from the same distribution (both p-values $>0.25$$>0.25$; two-sided alternative hypothesis: the locations are not shifted). The very high Spearman correlation coefficients ($0.879$$0.879$ and $0.935$$0.935$ for coverage and regularity, respectively) and very low MAEs ($0.021$$0.021$ and $0.052$$0.052$ for coverage and regularity, respectively) confirm that the two variants of the metrics are interchangeable.

This finding quantitatively support the equivalent qualitative assumption from Shmalts 2026.

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	cvg	reg
W	598.500	665.500
p	0.262	0.785
r	0.879	0.935
mae	0.021	0.052

3. Constraint Applicability

Description

This chapter is the core part of the notebook. It analyses the applicability values of the constraints, which includes their distribution and correlation, and seeks to establish a relation between the values of coverage and regularity of the constraints, and their ontological properties considered in chapter 1.

1. First, we qualitatively analyze the distribution of the applicability values of the constraints. We observe a natural clustering of the item coverage and regularity values, and propose a new (ir)regularity scale based on it.
2. We compare the item coverage and regularity values of the P2L vs. L2P constraints. The L2P constraints appear to be statistically more regular. The difference in coverage is unexpectedly insignificant.
3. We test for a correlation between coverage and regularity. We find that there exists a small negative correlation between both item and type coverage and regularity.
4. We investigate the effect of the domain type of constraints on their item regularity. We find that semantics is the only domain that has a significant effect on item regularity. As expected, the effect is negative.

NB! In this chapted, we filter out all unconfirmed constraints (item regularity of $<0.5$$<0.5$) and constraints with $<3$$<3$ covered items.

Results

- Most of the item coverage values are very low at $<0.05$$<0.05$. The item regularity values are more even. They are clustered naturally by the thresholds of $0.7$$0.7$ and $0.9$$0.9$, which appear to cut through the first and the third largest gaps between two neighboring item regularity values. We propose a new (ir)regularity scale of $[0.5; 0.7)$$[0.5; 0.7)$, $[0.7; 0.9)$$[0.7; 0.9)$, $[0.9; 1.0]$$[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints, while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho $-0.184$$-0.184$ and $-0.342$$-0.342$, respectively).
- We find that including semantic conditions to constraint definitions statistically sinks the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

First, we only keep constraints that have at least $3$$3$ covered items and an item regularity of $\geq 0.5$$\geq 0.5$. That results in $49$$49$ constraints. Again, the number of the P2L constraints doubles the number of the L2P constraints.

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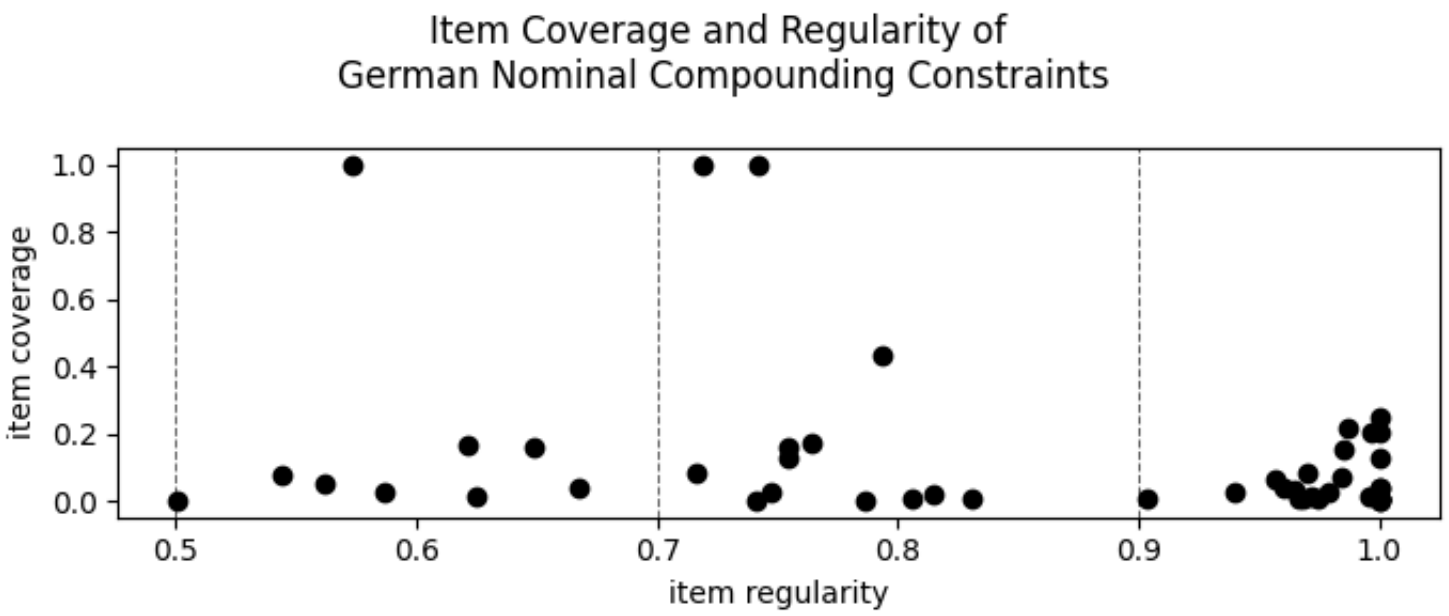
	N
total	49
p2l	32
l2p	16

1. As already reported in Shmalts 2026, the item coverage values are mostly very low, with more than half of the constraints lying in the item coverage range of $[\sim 0.0; 0.05)$$[\sim 0.0; 0.05)$ (28 of 49).

What concerns the item regularity values, the present study produces novel results. Specifically, it seems that the thresholds of $0.7$$0.7$ and $0.9$$0.9$ are the natural barriers between "low", "moderate", and "high" regularity. Indeed, the respective clusters are seen clearly on the scatter plot. The legitimacy of this values is further supported by the fact that they cut through the first and the second largest gaps between neighboring item regularity values in the dataset.

The item regularity values clustering calls the original regularity scale from Shmalts 2026 into question. We propose the newly discovered thresholds of $0.7$$0.7$ and $0.9$$0.9$ as the new (ir)regularity scale. More on completion of the new scale with a "regular" level follows in chapter 4.

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	N
cvg: [0.0-0.05)	28
cvg: [0.05-0.2)	13
cvg: [0.2-1.01)	8
reg: [0.5-0.7)	9
reg: [0.7-0.9)	13
reg: [0.9-1.01)	27

The two largest gaps betwee neighboring item regularity values are $0.831$$0.831$-$0.903$$0.903$ and $0.667$$0.667$-$0.716$$0.716$. They are cut precisely by the thresholds of $0.9$$0.9$ and $0.7$$0.7$, respectively.

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	left	right	diff
0	0.831	0.903	0.072
1	0.667	0.716	0.049
2	0.501	0.544	0.043
3	0.903	0.940	0.037
4	0.587	0.621	0.034
5	0.625	0.649	0.024
6	0.719	0.741	0.022
7	0.764	0.786	0.022
8	0.544	0.562	0.018
9	0.649	0.667	0.018

2. The L2P constraints appear to be statistically more regular than the P2L constraints. This conclusion corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirm its robustness after adding semantic constraints. On the contrary, there was no observed statistical difference between the item coverage values of the P2L vs. the L2P constraints, as unexpected as it was. This finding contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.

Almost all L2P applicability values have a very low coverage and a high regularity, while the applicability values of the P2L constraints are distributed more or less evenly. Thus, 12 out of 16, or $75\%$$75\%$ of the L2P constraints have an item coverage $<0.05$$<0.05$. For the P2L constraints, this number is only $50\%$$50\%$. The item regularity values are concentrated much closer to $1.0$$1.0$ among the L2P but not the P2L constraints. There are none L2P constraints with item regularity $<0.7$$<0.7$. On the contrary, there are eighr, or $25\%$$25\%$, such P2L constraints. Almost all L2P constraints are highly regular ($\geq 0.9$$\geq 0.9$): 13 out of 16, or $81.3\%$$81.3\%$. The P2L constraints do not even come close with mere $43.8\%$$43.8\%$ of highly regular constraints.

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	all	P2L	L2P
cvg: [0.0-0.05)	28	16	12
cvg: [0.05-0.2)	13	11	2
cvg: [0.2-1.01)	8	5	2
reg: [0.5-0.7)	9	8	0
reg: [0.7-0.9)	13	10	3
reg: [0.9-1.01)	27	14	13

We now run a Mann-Whitney U-test over the applicability values of the P2L vs. the L2P constraints. We choose the Mann-Whitney U-test because we are dealing with two independent samples whose distributions are abnormal and skewed. Since we suppose that the L2P constraints have lower coverage and higher regularity, we choose the two one-tailed alternative hypotheses: greater and less.

We can clearly see that the L2P constraints are undeniably more regular than the P2L constraints at sig = 0.001 (p-value $0.001$$0.001$; one-tailed alternative: L2P item regularity values are greater). They also have a lower coverage, with p-value of $0.017$$0.017$ (one-tailed alternative: P2L item coverage values are greater).

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		cvg	reg
alternative			
P2L>L2P	U	353.500	115.000
	p	0.017	0.999
P2L<L2P	U	353.500	115.000
	p	0.984	0.001

Since there are a few outliers in the P2L item coverage values, and *SP18_Pl_Constr* is a small dataset that can be not representative as of coverage, the results of the Mann-Whitney U-test can biased. We therefore additionally run a permutation test below. It compares the values themselves and not their ranks and is therefore much more sensitive to the outliers.

The permutation test confirmed that the item regularity values are higher among the L2P constraints (observed shift of the P2L vs. the L2P item regularity means $-0.131$$-0.131$, p-value $0.001$$0.001$). It also confirms our doubts about item coverage: the difference is not statistically different, with p-value of $\sim 0.06$$\sim 0.06$.

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		cvg	reg
alternative			
P2L>L2P	shift	0.088	-0.131
	p	0.057	0.998
P2L<L2P	shift	0.088	-0.131
	p	0.943	0.002

To demonstrate the bias of the outliers, we run a dedicated Mann-Whitney U-test over *GeCoDB_v07_GN* only with an item coverage threshold og $0.5$$0.5$, which is still much. While the robust regularity difference stays significant at p-value $0.003$$0.003$, the significance of the coverage difference vanishes (p-value $0.072$$0.072$). We conclude that there is no difference in item coverage between the P2L and the L2P constraints, after all.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

		cvg	reg
alternative			
P2L>L2P	U	235.500	89.000
	p	0.072	0.997
P2L<L2P	U	235.500	89.000
	p	0.932	0.003

3. Contradictory to the claim in Shmalts 2026, there exists a small negative correlation between both item and type coverage and regularity (Spearman rho $-0.184$$-0.184$ and $-0.342$$-0.342$, respectively). Statistically, constraints with lower coverage have slightly higher regularity. This is in line with our assumption that the more specific constraints are stronger than the general ones. The large difference in the correlation coefficient between the item and type variants is a result of different productivity of items what regards building compounds.

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	item	type
r	-0.184	-0.342

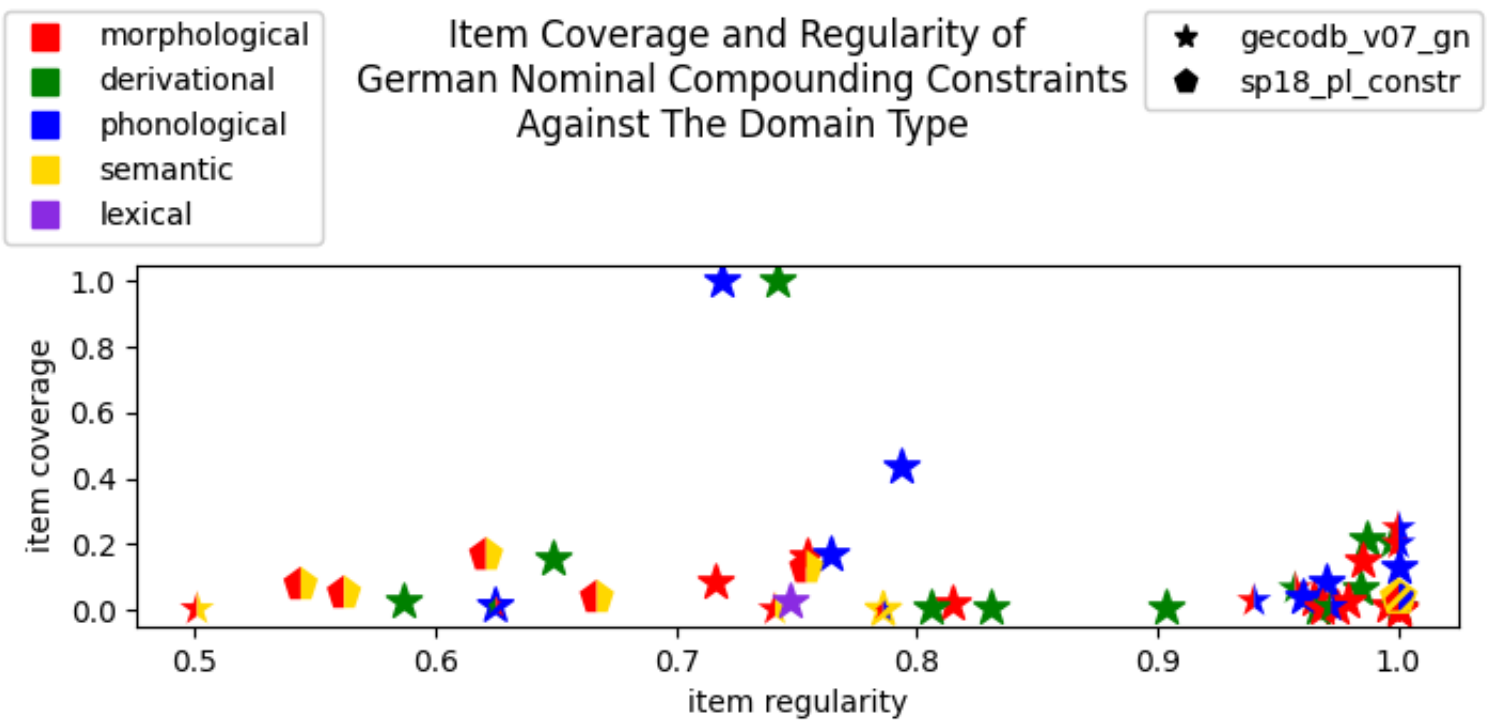
4. We find that including semantic conditions to constraint definitions statistically sinks the item regularity of the constraints (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding. Note that all non-confirmed constraints (item regularity $<0.5$$<0.5$) are semantic as well which makes the real picture even worse for semantics.

At this step, we will be treating the domain type of constraints as one-hot vectors. For example, a morphological+phonological constraint will be encoded as $[1, 0, 1, 0, 0]$$[1, 0, 1, 0, 0]$, which is decoded as `[+morphological][+derivational][+phonological][+semantic][+lexical]`. These vectors will then be passed to a linear regression model with an Ordinary Least Squares (OLS) loss function.

Here, we first create the vectors.

Let us consider the scatter plot of constraint item regularity against the domain type. We can immediately see that morphological, derivational, and phonological constraints are scattered all over the lot but are still concentrated towards the right edge of the axis. Contrary to that, semantic (including mixed) constraints only populate the range of $[0.5; 0.8]$$[0.5; 0.8]$; the only constraint with a high regularity is morphological+phonological+semantic. The plot suggests that the constraints that include a semantic condition are less regular.

```
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```



To confirm the intuition, we run a linear regression over the item regularity of the constraints vs. their domain type, based on the one-hot vectors we produced above. To concentrate on individual contribution of each domain, we do not include their interaction terms.

The linear regression model confirms our intuition. Semantics is the only domain that contributes to item regularity significantly (coeff $-0.21$$-0.21$ at intercept $0.85$$0.85$, p-value $0.001$$0.001$). That is, the item regularity of an arbitrary constraint sinks by $0.21$$0.21$ when a semantic condition included, assuming that the other domains of this constraint remain intact. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.289
Model:                            OLS   Adj. R-squared:                  0.205
Method:                 Least Squares   F-statistic:                     3.418
Date:                Thu, 06 Aug 2026   Prob (F-statistic):             0.0111
Time:                        16:08:06   Log-Likelihood:                 30.094
No. Observations:                  48   AIC:                            -48.19
Df Residuals:                      42   BIC:                            -36.96
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept                  0.8499      0.061     13.936      0.000       0.727       0.973
morphological[T.True]      0.0643      0.058      1.109      0.274      -0.053       0.181
derivational[T.True]   -6.965e-05      0.058     -0.001      0.999      -0.116       0.116
phonological[T.True]       0.0540      0.048      1.122      0.268      -0.043       0.151
semantic[T.True]          -0.2074      0.055     -3.742      0.001      -0.319      -0.096
lexical[T.True]           -0.1029      0.151     -0.681      0.499      -0.408       0.202
==============================================================================
Omnibus:                        2.435   Durbin-Watson:                   1.479
Prob(Omnibus):                  0.296   Jarque-Bera (JB):                1.949
Skew:                          -0.494   Prob(JB):                        0.377
Kurtosis:                       2.990   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

If we include the interaction terms, the picture stays the same. Now morphology, derivation, phonology, and semantics (but not the lexical domain) significantly contribute to item regularity (p-value in $[0.0; 0.001]$$[0.0; 0.001]$). From these four isolated domains, only semantics has a negative coefficient ($-0.128$$-0.128$) which the former three increase item regularity by $0.274$$0.274$, $0.174$$0.174$, $0.196$$0.196$, respectively (intercept $0.672$$0.672$). From all the interaction terms, the interaction of morphology and semantics is significant (coefficient $-0.128$$-0.128$, p-value $0.0$$0.0$). Otherwise, the interaction morphology+derivation+phonology is significant, too (coefficient $-0.352$$-0.352$, p-value $0.034$$0.034$) but we discard it since there is only one such constraint, which cannot assure statistical power.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.469
Model:                            OLS   Adj. R-squared:                  0.344
Method:                 Least Squares   F-statistic:                     3.734
Date:                Thu, 06 Aug 2026   Prob (F-statistic):            0.00195
Time:                        16:08:06   Log-Likelihood:                 37.106
No. Observations:                  48   AIC:                            -54.21
Df Residuals:                      38   BIC:                            -35.50
Df Model:                           9
Covariance Type:            nonrobust
================================================================================================================
                                                                                                        c
----------------------------------------------------------------------------------------------------------------
Intercept                                                                                              0.6
morphological[T.True]                                                                                  0.2
derivational[T.True]                                                                                   0.1
phonological[T.True]                                                                                   0.1
semantic[T.True]                                                                                      -0.1
lexical[T.True]                                                                                        0.0
morphological[T.True]:derivational[T.True]                                                            -0.1
morphological[T.True]:phonological[T.True]                                                            -0.1
derivational[T.True]:phonological[T.True]                                                             -0.0
morphological[T.True]:semantic[T.True]                                                                -0.1
derivational[T.True]:semantic[T.True]
phonological[T.True]:semantic[T.True]                                                                  0.0
morphological[T.True]:lexical[T.True]
derivational[T.True]:lexical[T.True]
phonological[T.True]:lexical[T.True]
semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]                                       -0.3
morphological[T.True]:derivational[T.True]:semantic[T.True]
morphological[T.True]:phonological[T.True]:semantic[T.True]                                            0.0
derivational[T.True]:phonological[T.True]:semantic[T.True]
morphological[T.True]:derivational[T.True]:lexical[T.True]
morphological[T.True]:phonological[T.True]:lexical[T.True]
derivational[T.True]:phonological[T.True]:lexical[T.True]
morphological[T.True]:semantic[T.True]:lexical[T.True]
derivational[T.True]:semantic[T.True]:lexical[T.True]
phonological[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:semantic[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
derivational[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
morphological[T.True]:derivational[T.True]:phonological[T.True]:semantic[T.True]:lexical[T.True]
==============================================================================
Omnibus:                        0.912   Durbin-Watson:                   1.901
Prob(Omnibus):                  0.634   Jarque-Bera (JB):                0.275
Skew:                          -0.055   Prob(JB):                        0.872
Kurtosis:                       3.354   Cond. No.                     1.46e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 3.97e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

The domain tyoe of constraints has no effect on their item coverage.

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```
                            OLS Regression Results
==============================================================================
Dep. Variable:                cvg_item   R-squared:                       0.117
Model:                             OLS   Adj. R-squared:                  0.012
Method:                  Least Squares   F-statistic:                     1.110
Date:                 Thu, 06 Aug 2026   Prob (F-statistic):              0.370
Time:                         16:08:06   Log-Likelihood:                 10.975
No. Observations:                   48   AIC:                            -9.951
Df Residuals:                       42   BIC:                             1.277
Df Model:                            5
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
Intercept                   0.2433      0.091      2.678      0.011       0.060       0.427
morphological[T.True]      -0.1578      0.086     -1.829      0.075      -0.332       0.016
derivational[T.True]       -0.1036      0.086     -1.208      0.234      -0.277       0.069
phonological[T.True]        0.0132      0.072      0.184      0.855      -0.131       0.158
semantic[T.True]           -0.0338      0.083     -0.409      0.684      -0.200       0.133
lexical[T.True]            -0.2173      0.225     -0.966      0.340      -0.671       0.237
==============================================================================
Omnibus:                       54.471   Durbin-Watson:                   0.802
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              276.619
Skew:                           3.002   Prob(JB):                     8.57e-61
Kurtosis:                      13.113   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

4. Constraint Support, Productivity & Regularity

Description

In this section, we introduce the new (irregularity) scale and apply it to our results.

1. We analyze the productivity, regularity, and support level of the constraints. Most of the constraints are productive and irregular to a different extent.
2. We that manually analyze partially or not supported constraints and find that no constraints require changes in their conditions.

NB! In this chapter, we take a look at the entire set of constraints, whether they are confirmed or not.

Results

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 are now partially supported. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*.
- Eleven constraints out of 49 turn out to be unproductive by the Tolerance Principle. Regular constraints by the RTP constitute slightly less than a half of the constraint set: 23 out of 49.
- We find that no constraints require changes in their conditions. Only regularity levels should be adjusted for them.

Yang 2016 formulates the Tolerance Principle that describes the maximum number of exceptions that a productive rule (in a broader meaning of a linguistic generalization) can tolerate:

$$ {\theta}_N < \frac{N}{ln N} $$$$ {\theta}_N < \frac{N}{ln N} $$

where $N$$N$ is the number of items that a rule is potentially applicable to, and ${\theta}_N$${\theta}_N$ is the maximum tolerable number of exceptions. Each rule, be it productive or not, is defined over a subset of items with certain linguistic properties, identically to our constraints. Thus, the whole relevant vocabulary is partitioned into subsets with certain linguistic properties, and each subset has its own size $N$$N$ and its own (un)productive rule defined over it. The crucial difference between productive and unproductive rules is then, that productive rules are applied automatically to any novel input that satisfies certain linguistic conditions, while unproductive rules are "frozen" and are bound to a certain set of items. Yang 2016 provides numerous evidence from psycholinguistic experiments of language acquisition in support of this distinction. Thus, children cross-linguistically readily overgeneralize productive but virtually never unproductive rules.

Yang 2016 claims that linguistic phenomena are governed by sets of productive and unproductive rules that are applied following thr Elsewhere Condition. Thus, the item input is first tested against the target linguistic properties of the first rule in the set. If the conditions are satisfied, the rule is applied and the derivation stops. If not, the conditions of the second rules applied and so on, until a matching rule is found or until the derivation reaches the last, least specific, default rule. On the example of German plural compounding, Yang 2016 demonstrates that German nouns can be partitioned such that each portion has a productive plural formation rule as tested against the Tolerance Principle. Thus, feminine nouns productively build a plural with *-en*, (non-feminine) nouns ending in schwa as well, then (non-feminine) nouns with condition X build a plural with y, etc. The distinction between productive and unproductive rules is important insofar that the unproductive rules come first in the Elsewhere Condition. How the unproductive and productive rules are ordered between themselves is an open question. Yang 2016 support this hypothesis with evidence from psycholinguistic experiments, too.

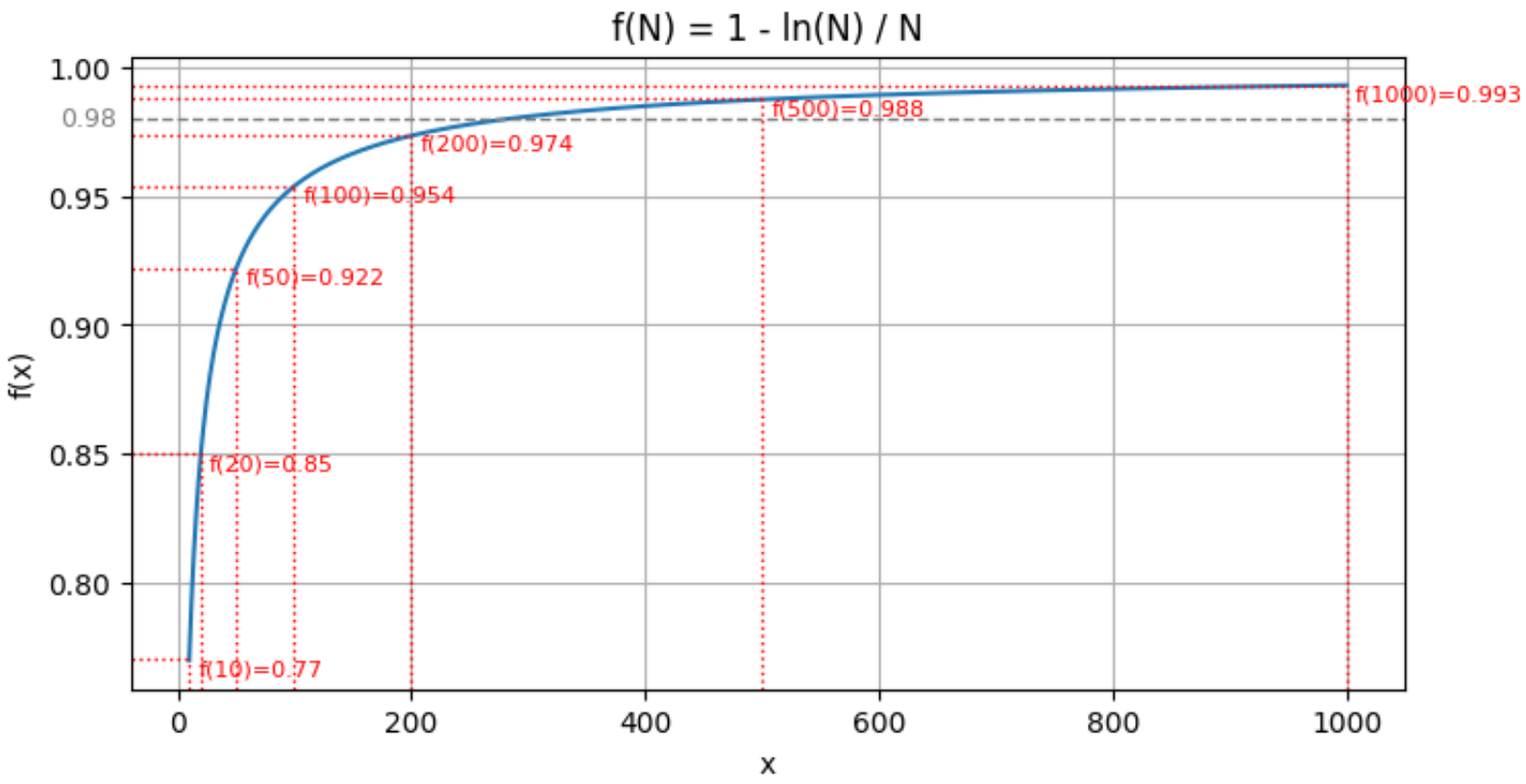
While we will try to model German nominal compounding with Yang's method, it will not be part of this study. This introduction into the Tolerance Principle was given to justify its validity or at least utility. We employ this numeric expression to derive a flexible constraint regularity threshold that accounts for its absolute item coverage. We find that a constant value may become unreasonable when the absolute item coverage is either too low or too high. In the original study, we choose $0.98$$0.98$ as the threshold for regularity. However, if we fit this threshold onto a rule that is potentially applicable to 3000 items, the rule will still be considered regular despite having 60 exceptions which is a lot for a generalization that should be applicable to virtually any input. On the other hand, there are a lot of constraints with an absolute coverage of 20-30 constraints. At this magnitude, $0.98$$0.98$ doesn't even make sense. We find that the multiplicative inverse of the Tolerance Principle threshold is a gut estimate of the maximum acceptable share of violations for a regular constraint. We propose the Regularity Tolerance Principle: a constraint (rule in Yang's terminology) is regular if for its item regularity $\text{reg}$$\text{reg}$ it holds:

$$ \text{reg} \geq 1 - \frac{1}{{\theta}_N} = 1 - \frac{ln N}{N} $$$$ \text{reg} \geq 1 - \frac{1}{{\theta}_N} = 1 - \frac{ln N}{N} $$

On another note, it is not quite clear how to distinguish actual grammatical constraints from mere statistical patterns. Maybe the Tolerance Principle may offer its service here.

Below is the visualization of the Regular Tolerance Principle and thresholds for different absolute item coverage values.

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	TP	ITP
10	4	0.770
20	6	0.850
50	12	0.922
100	21	0.954
200	37	0.974
500	80	0.988
1000	144	0.993
5000	587	0.998

Based on the results of chapter 3 and the Regularity Tolerance Principle, we propose a new (ir)regularity scale. The scale has two layers. The first layer distinguishes regular from irregular constraints, governed by the Regularity Tolerance Principle (RTP). If the constraint is irregular, we define it (ir)regularity level by the thresholds of $0.7$$0.7$ and $0.9$$0.9$. The layered scale is required to account for small-scale constraints, for which the regularity threshold by the RTP may be below $0.9$$0.9$. The procedure of determining the support level of constraints remains that same as in the original study. Reminder: we declare any constraints with an item regularity of $<0.5$$<0.5$ not supported.

1. The new (ir)regularity scale did not affect the support level of previously verified constraints much. Only four constraints from Shmalts 2026 are now partially supported. The newer versions of *GeCoDB* are even better with three and two partially supported constraints. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*. Note that some of the constraints had zero coverage in all their datasets.

Eleven constraints out of 49 turn out to be unproductive by the Tolerance Principle. Regular constraints by the RTP constitute slightly less than a half of the constraint set: 23 out of 49.

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	gecodb_v06	gecodb_v07	gecodb_v07_gn	sp18_pl_constr
supported+	12	12	12	4
supported	23	24	28	4
partially supported	4	3	2	0
not supported	0	0	2	2

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is_productive	is_regular	N
False	False	11
True	False	13
	True	24